

NewsLens AI Project Report

Synthetic-only classifier, calibration, evaluation and deployment evidence

All classifier training articles and entities are synthetic. This report documents a bounded ledger-consistency demonstration, not unrestricted real-world fact verification.

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Document control

Field	Value
Project	NewsLens AI
Phase	5S — synthetic dataset, model and deployment
Author	Deven Sachin Gaikwad
Status	PR B evidence package; external deployment values recorded after protected merge

NewsLens AI

NewsLens AI is a noncommercial student and placement-portfolio demonstration built by **Deven Sachin Gaikwad**. It combines deterministic extractive summarisation with an explainable classifier trained exclusively on an independently created synthetic benchmark.

The classifier has a deliberately narrow purpose: compare the visible Reference note and Article account fields in the supported fictional format and indicate whether their ledgers are consistent or contradicting. It is **not** a general fake-news detector and does not establish real-world truth.

What is public

- 24,000 original synthetic articles representing 12,000 fictional paired events.
- Deterministic generator, manifest, audit evidence, and reproducibility tests.
- TF-IDF bigram + Logistic Regression public model.
- Platt calibration cryptographically bound to the exact model SHA-256.
- Group-safe training, validation, calibration, abstention-policy, and locked-test partitions.
- Streamlit interface with summarisation, classification, abstention, explanations, exports, session-local review, and aggregate monitoring.

No ISOT row, copied phrase, close paraphrase, transformation, translation, summary, reconstruction, private ISOT model, private calibration artifact, or external copyrighted training dataset is included.

Measured evidence

Measure	Result
Locked final-test rows	1,200
Accuracy / balanced accuracy / macro F1	1.000 / 1.000 / 1.000
Confusion matrix	`[[600, 0], [0, 600]]`
Platt Brier score	0.000001497
Expected calibration error	0.000742
Counterfactual account-swap flip rate	100%
Surface-text-only balanced accuracy	0.500
Fact-block ablation balanced accuracy	0.498
Metadata-only balanced accuracy	0.521

Perfect in-distribution performance reflects a structured synthetic comparison task; it must not be interpreted as unrestricted news-veracity performance.

Run locally

Python 3.12 is the deployment target.

```
python -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements-lite.txt
python -m streamlit run app.py
```

On Windows PowerShell, activate with `.venv\Scripts\Activate.ps1`.

No API key or secret is required. The packaged model and calibration are loaded from `models/` without runtime training.

Reproduce the model package

The authoritative dataset archive is already versioned. The following command verifies its identity, trains on the authored training split, selects only on model validation, fits calibration, selects the abstention policy, and evaluates the locked final test once:

```
python training/train_synthetic_model.py
```

The accepted public artifacts are described in `models/public_artifact_manifest.json`. Re-running training in a materially different dependency environment can change serialized bytes; publication uses the recorded artifacts and hashes.

Validate

```
python -m compileall -q app.py pages src ui tests scripts training synthetic_benchmark
python -m pytest -q --strict-markers
python scripts/audit_public_release.py --allow-publication-gates
```

The release scan verifies the dataset ZIP, public model, calibration binding, legal package, secrets, private artifact exclusions, navigation, and local links.

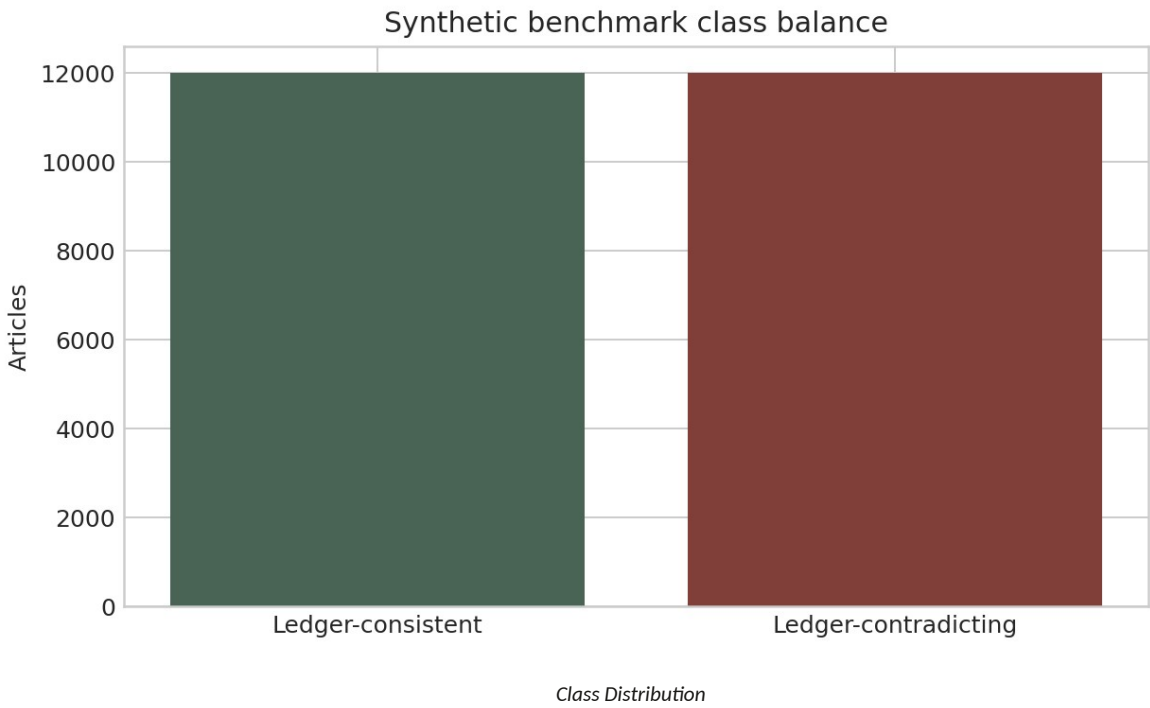
Responsible use

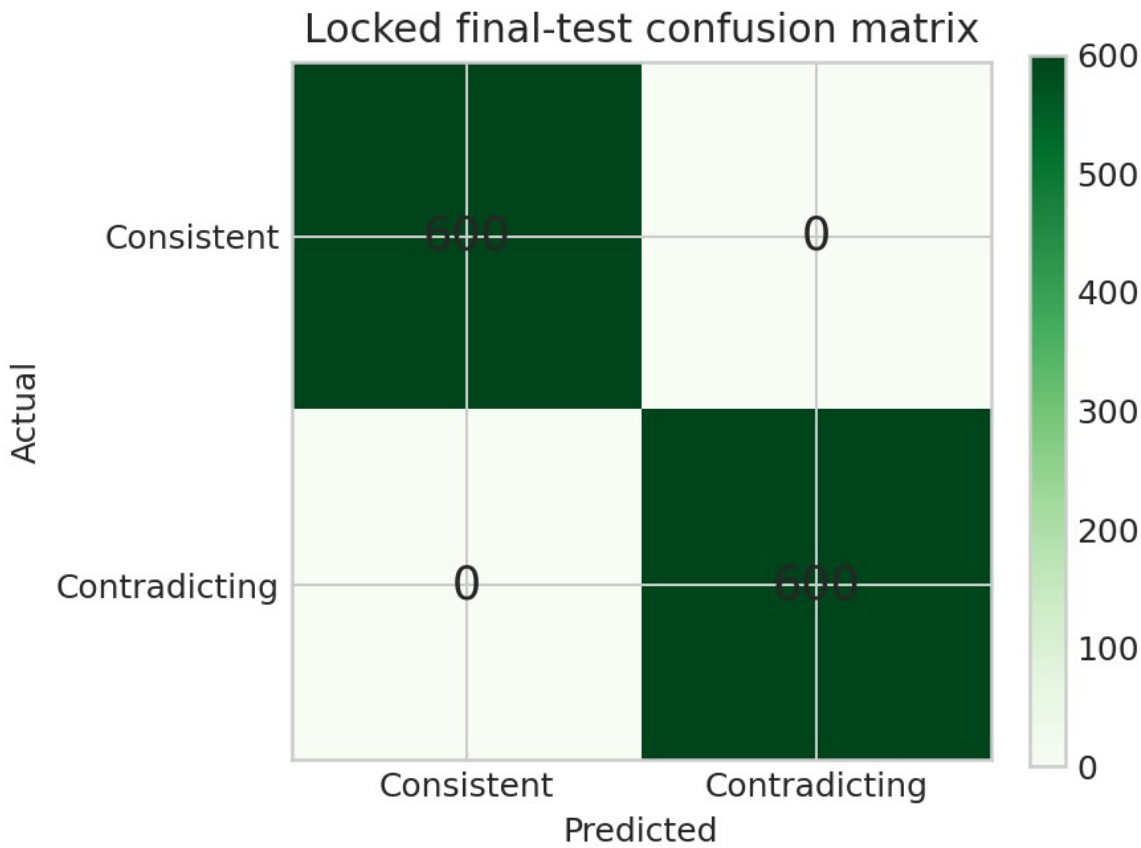
- Inputs lacking both supported fact blocks are routed to **Editorial review required**.
- Confidence measures agreement with synthetic benchmark labels, not factual truth.
- The app does not expose raw training rows.
- Public history is isolated to a visitor session; full article text is not persisted.
- Human verification against independent primary sources remains necessary.

See `docs/DATASET_CARD.md`, `docs/MODEL_CARD.md`, `docs/TESTING.md`, and `docs/DEPLOYMENT.md` for the full evidence trail.

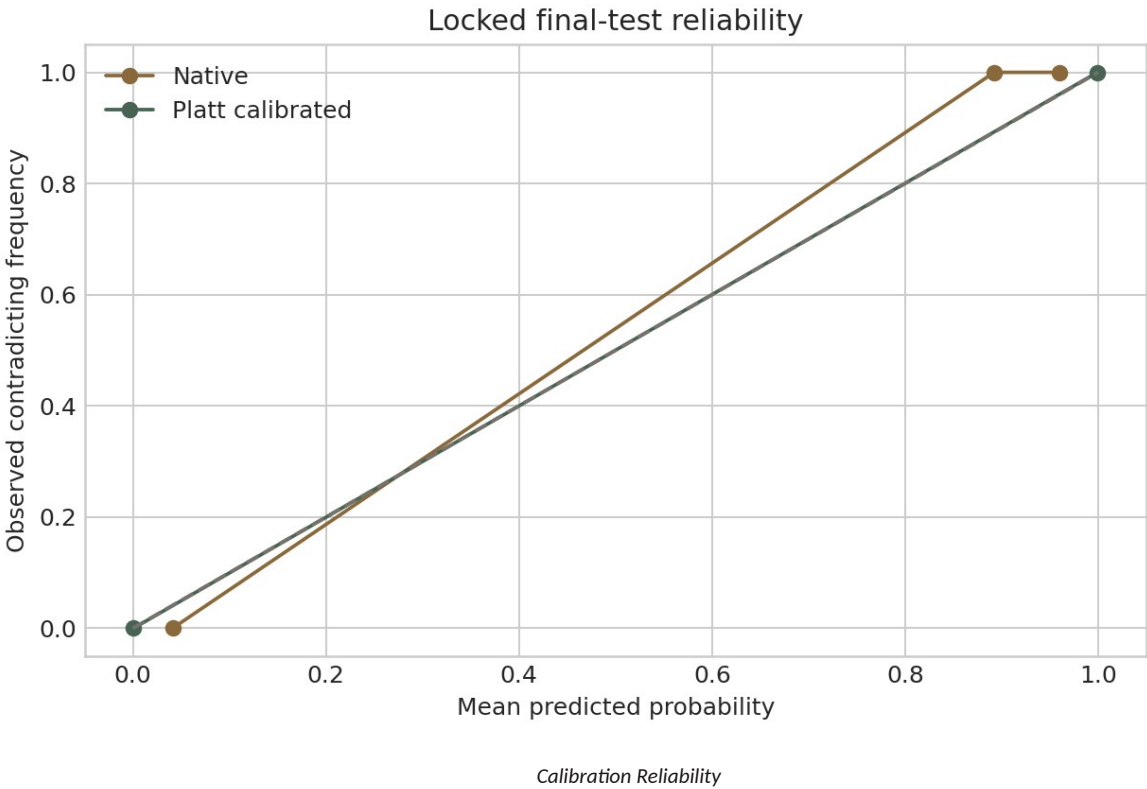
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Selected measured figures





Confusion Matrix



NewsLens AI Phase 5S Deployment and Audit Report

Updated: 26 September 2026

Dataset release (PR A)

- PR: #44
- URL: <https://github.com/DevenGaikwad/NewsLens-AI/pull/44>
- Owner upload commit: e6031fa6ec193131be5fed4704722125104b066f
- Follow-up head: 326499bb6f58f0420a0cf36a00828a901836323d
- Squash merge: d3e77e6610c4a9c83d9137ca672a9aad4278285f
- Resulting tree: 520bad55ea12f1a8e48ad2e533af26312c62933d
- Signature: GitHub verified
- Merge time: 2026-09-25T12:14:59Z
- Required PR and exact-new-main checks: passed

Dataset identity

- Path: data/synthetic/newslens-synthetic-articles-v1.0.0.zip
- Size: 10,319,934 bytes
- SHA-256: 3b6df1fa17615bfe1b67f6c9136909c668ec846e3fa8d205fa8e4aa2a80526cc
- Git blob: cb0d5e57407be10fecefb34762e586bacd38dd56
- License: CC BY 4.0 to the extent applicable rights subsist
- Provenance: independently generated original fictional articles and entities

Model evidence

- Model: models/newslens_synthetic_pipeline.joblib
- Model SHA-256: c1ad8c044cd95bc7bf25a94716010ddbe21fbae2ec92a0c1cefb01f5c3c979c6
- Calibration: models/newslens_synthetic_calibration.json
- Calibration SHA-256: adcf03a860ef8fb41058b3a4dcc80351dbd02e05f81e0fc1e7f971624af6ab77
- Binding: exact model hash recorded in calibration and public manifest
- Training / validation / calibration / policy / test rows: 18,000 / 2,400 / 1,200 / 1,200 / 1,200
- Cross-partition event/content overlap: 0 / 0
- Locked accuracy, balanced accuracy, macro F1: 1.0 / 1.0 / 1.0
- Brier / ECE: 0.000001497 / 0.000742
- Coverage at threshold 0.50: 100%
- Counterfactual flip rate: 100%
- Surface-only / ablation / metadata balanced accuracy: 0.500 / 0.498 / 0.521

Publication boundary

No ISOT content, private ISOT-derived artifact, or external copyrighted article dataset is in the model, repository archive, or intended release. Noncommercial status was not used as a substitute for rights.

PR B and deployment

PR B number, validated head, squash merge, resulting tree, exact-new-main workflow evidence, Streamlit URL, deployed commit, and public smoke-test evidence are pending execution. These fields will be updated after the protected merge and live deployment; they are not claimed in advance.

Vercel and six retained Dependabot PRs remain outside scope.

Editorial AI Case Study: From Blocked Artifact to Synthetic-Only System

Problem

The original demonstration depended on a private classifier whose training-data redistribution position did not support public deployment. Publishing source without a working model would create a misleading portfolio artifact; publishing the private artifact would violate the project's evidence and rights policy.

Intervention

NewsLens AI introduced an independently authored fictional benchmark of 24,000 articles / 12,000 paired events. Each article exposes a Reference note and Article account. Consistent examples repeat the authored facts; contradicting examples alter one or more fields. Event-grouped partitions prevent paired variants from crossing training and evaluation boundaries.

Three candidates were compared on model validation. A compact Logistic Regression pipeline was retained, Platt calibration was fitted on a separate partition, and the review threshold was selected on another partition. The final test remained locked until all decisions were fixed.

Results

The locked test achieved 1.0 accuracy, balanced accuracy, and macro F1 with a $\begin{bmatrix} 600 & 0 \\ 0 & 600 \end{bmatrix}$ confusion matrix. Brier score was 0.000001497 and ECE 0.000742. Surface text and fact-block ablation remained at chance, metadata was near chance, and swapping only Article account values flipped every paired prediction.

Product integration

The public model and calibration are hash-bound. Missing structured blocks force human review. Deterministic summarisation remains separate from classification. The app preserves explanations, exports, privacy-safe session history, responsive presentation, and honest limitations without an external checkpoint or paid API.

Lesson

The central engineering achievement is not a claim of universal fake-news detection. It is a traceable replacement of an unpublishable dependency with a bounded, reproducible, original, deployable system whose success criteria match its declared task.

Licensing Status

Current public package

NewsLens AI is a proprietary, source-visible student and portfolio demonstration. Original application code, interface materials, reports, and documentation are © 2026 Deven Sachin Gaikwad, All Rights Reserved, unless a file expressly states otherwise.

The independently created newslens-synthetic-articles-v1.0.0 dataset is offered under CC BY 4.0 to the extent applicable rights subsist. Its generator, provenance, and identity are recorded in docs/DATASET_CARD.md and data/synthetic/.

The public model and calibration were created exclusively from that dataset and are intentionally distributed with this demonstration under the repository's stated source-visible terms. Third-party Python and JavaScript packages retain their own licenses.

Excluded material

No external copyrighted article dataset is redistributed. No ISOT row, phrase transformation, private classifier, calibration, vocabulary, coefficient, prediction, or evaluation artifact is present. The earlier private-artifact redistribution question is resolved by replacement, not by assuming permission.

Noncommercial educational use is not treated as automatic copyright permission.